

Lecture 21 — Systems and $Ax = b$

Mon Oct 12

Last updated: August 18, 2026.

The class outline for this day will be posted before class.

Related reading and the course-wide list of daily objectives: [Lectures](#).

Learning objectives

By the end of class, students should be able to:

1. Translate equations or contextual constraints into $Ax = b$.
2. Interpret $Ax = b$ as asking whether b belongs to the span of the columns of A .
3. Classify systems as square, underdetermined, or overdetermined.
4. Anticipate circumstances producing zero, one, or many solutions.